

CorrMatch: Label Propagation via Correlation Matching for Semi-Supervised Semantic Segmentation

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Abstract

This paper presents a simple but performant semi-supervised semantic segmentation approach, called *CorrMatch*. Previous approaches mostly employ complicated training strategies to leverage unlabeled data but overlook the role of correlation maps in modeling the relationships between pairs of locations. We observe that the correlation maps not only enable clustering pixels of the same category easily but also contain good shape information, which previous works have omitted. Motivated by these, we aim to improve the use efficiency of unlabeled data by designing two novel label propagation strategies. First, we propose to conduct pixel propagation by modeling the pairwise similarities of pixels to spread the high-confidence pixels and dig out more. Then, we perform region propagation to enhance the pseudo labels with accurate class-agnostic masks extracted from the correlation maps. *CorrMatch* achieves great performance on popular segmentation benchmarks. Taking the DeepLabV3+ with ResNet-101 backbone as our segmentation model, we receive a 76%+ mIoU score on the Pascal VOC 2012 dataset with only 92 annotated images. Code is available at <https://github.com/BBBBchan/CorrMatch>.

1. Introduction

With the development of deep learning techniques, especially convolutional neural networks (CNNs) [12, 14, 21, 55, 60, 69], many significant semantic segmentation methods [5, 15, 18, 28, 65, 68, 71] have achieved remarkable results. However, methods based on deep learning often require large-scale pixel-wise annotated datasets with a massive amount of labeled images. Compared to the image classification and object detection tasks [8, 38], the accurate annotations for segmentation datasets are very expensive and time-consuming.

Recently, many researchers have sought to address the above challenge by reducing the demand for large-scale

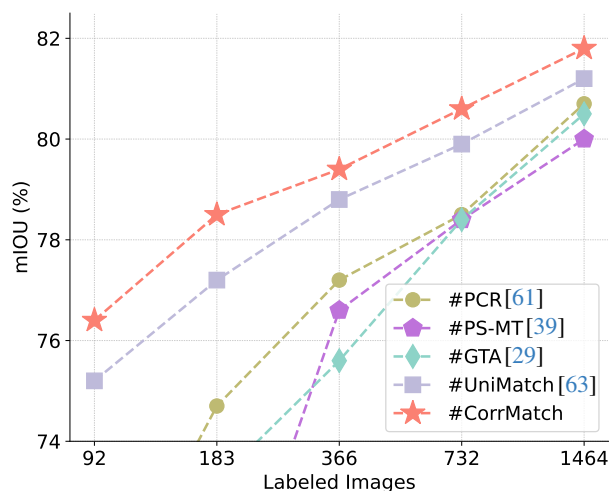


Figure 1. Comparison with state-of-the-art methods on the Pascal VOC dataset. Our CorrMatch outperforms all others for all splits.

accurately annotated data in the semantic segmentation task by presenting weakly-supervised [26, 27, 53, 56], semi-supervised [11, 22, 23, 41], or even unsupervised segmentation methods [13, 19, 24, 50]. Among these schemes, semi-supervised semantic segmentation only requires a small amount of labeled data accompanied by a large amount of unlabeled data for training, which approaches real-world scenarios more and hence attracts the favor of more and more researchers from academia and industry.

In the literature of semi-supervised semantic segmentation, most works adopt the Mean Teacher architecture [23, 29, 39, 61] or self-training strategy [31, 64, 66] to enable consistency regularization. As shown in Tab. 1, these methods often require extra networks or training stages, complicating the training process. Although the recent UniMatch [63] has shown that a single-stage pipeline is sufficient, it still demands multiple strong augmentation data streams. Unlike them, our CorrMatch is a simpler framework with no need for multiple networks, training stages, or strong augmentation data streams.

Furthermore, in previous works [39, 61, 64], the most pop-

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Table 1. Differences between our CorrMatch and some representative approaches. SDA denotes strong data augmentation.

Method	Multiple networks	Multi-train stages	Multiple SDA streams	Pairwise similarity
PS-MT [39]	✓	✗	✗	✗
ST++ [64]	✗	✓	✗	✗
ELN [34]	✓	✓	✗	✗
UniMatch [63]	✗	✗	✓	✗
CorrMatch	✗	✗	✗	✓

ular way to leverage unlabeled data is setting a fixed threshold to screen reliable pixels as pseudo labels. However, those methods often struggle to efficiently utilize unlabeled data due to the trade-off between pseudo-label proportion and accuracy via threshold adjustments. Beyond that, motivated by the fact that the correlations between pixels can reflect the pairwise similarities, which indicates semantically similar pixels exhibit higher similarity on the correlation map, we reconsider the challenge of accurately assigning pseudo labels to unlabeled data from a label propagation perspective.

First, considering the correlation maps embed the global pairwise similarities, we propose the pixel propagation strategy. With correlation maps constructed from extracted features, the pixel propagation strategy spreads them into predictions, which enriches predictions with global similarities information and fosters semantic consistency. Meanwhile, with the observation that every row of a correlation map is equipped with local shape information, a series of binary maps that capture the objects' shapes can be acquired. Thus, coupled with the most salient predicted class within the intersection of the shapes and high-confidence regions, we propose the region propagation strategy to enhance pseudo labels by accurately assigning class labels to these shapes. By considering the union of shapes and high-confidence regions as the new ones, the high-confidence regions can be expanded, consequently improving the use efficiency of unlabeled data. As shown in Fig. 1, our CorrMatch outperforms all previous approaches.

Our main contributions can be summarized as follows:

- We demonstrate the two advantages of correlation maps in improving the use efficiency of unlabeled data.
- We design a simple but performant semi-supervised semantic segmentation framework containing two novel label propagation strategies.
- Our CorrMatch achieves new state-of-the-art performance on the Pascal VOC 2012 and Cityscapes datasets without any computation burden during inference.

2. Related Work

2.1. Semi-Supervised Learning

Semi-supervised learning [44, 76] is proposed to settle a paradigm that how to construct models using both labeled

and unlabeled data and has been studied long before the deep learning era [2, 3, 30]. And certainly, semi-supervised learning has gained more attention with advancements in deep learning and computer vision [4, 16, 37, 58, 59, 77].

Since Bachman *et al.* [1] proposed a consistency regularization-based method, many approaches, such as Π -Model [36, 43], Mean Teacher [48] and Dual Student [33] have migrated it into the semi-supervised learning field. Recently, FixMatch [46] provides a simple weak-to-strong consistency regularization framework and serves as many other relevant methods' baseline [17, 47, 49, 63]. However, many follow-up works [51, 62, 67] have pointed out that simply setting a manually fixed threshold may lead to inferior performance and slow convergence speed. Among them, FreeMatch [51] provides a dynamic threshold scheme connected with the model's learning process. However, these strategies designed for classification are not suitable for segmentation as multiple categories often exist in each image.

2.2. Semi-Supervised Semantic Segmentation

As semi-supervised learning has achieved surprising results in the image classification [36, 37, 46, 48], many works adopt the same setting for semantic segmentation [22, 41, 57].

One type of methods [11, 23, 39, 52, 61, 70, 72, 75] adopt the Mean Teacher architecture. U²PL [52] attempts to use unreliable predictions via contrastive learning better. PS-MT [39] builds a stricter teacher with the VAT [40] technique. ELN [34] uses an error localization network to mitigate the performance degradation caused by confirmation bias due to invalid pseudo labels. All of these methods demand multi-networks for training. Meanwhile, another type of method, self-training based methods [9, 31, 64, 66], often require multiple training stages. Among them, ST++ [64] proposes a three-stage paradigm with strong augmentation. SimpleBase [66] uses separated batch normalization [25] for images with different augmentation. PC²Seg [74] uses feature-space contrastive learning besides consistency training. Recently, UniMatch [63] adopted a single-stage framework based on FixMatch [46] via multiple strong augmentation branches. Unlike all the above, CorrMatch explores how to take advantage of correlation maps better to improve the use efficiency of unlabeled data via label propagation.

3. CorrMatch

The goal of semi-supervised semantic segmentation is to train a semantic segmentation network $\mathcal{F}$ with a small labeled image set and a large unlabeled image set. We present a single-stage framework CorrMatch, which leverages pairwise correlations to achieve two label propagation strategies.

3.1. Preliminaries

CorrMatch is built upon a simple framework [63] with weak-to-strong consistency regularization. A standard cross-

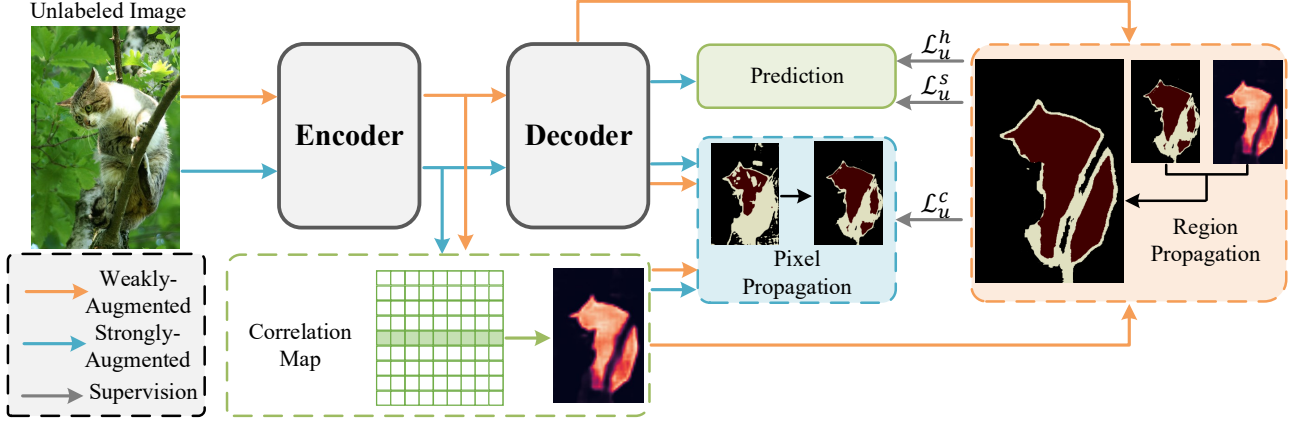


Figure 2. Illustration of our CorrMatch pipeline for unlabeled images. We build it upon the DeepLabv3+ framework [5]. Besides consistency regularization, CorrMatch adopts two label propagation strategies with correlation matching.

entropy loss is applied for labeled images $\{x_i^l\}$ and their corresponding labels $\{y_i^l\}$. And unlabeled images $\{x_i^u\}$ are mainly leveraged by enforcing prediction consistency. For an unlabeled image, x_i^w and x_i^s represent its augmented version with weak and strong augmentation, respectively. The consistency regularization treats the prediction of x_i^w as the pseudo label for x_i^s . We demonstrate the pipeline of unlabeled images in Fig. 2.

Given a mini-batch of N unlabeled images, we encourage the outputs to be consistent for both weakly and strongly augmented inputs with hard supervision:

$$\mathcal{L}_u^h = \frac{1}{N} \sum_i \ell_c(\mathcal{F}(x_i^s), \mathcal{F}(x_i^w)) \odot \mathcal{M}_i, \quad (1)$$

where ℓ_c is the pixel-wise cross-entropy loss function and $\odot$ is the element-wise multiplication. $\mathcal{M}_i$ is a binary map indicating the positions with high confidence predictions in $\mathcal{F}(x_i^w)$, which can be written as:

$$\mathcal{M}_i = \mathbb{1}(\max(\hat{\mathcal{F}}(x_i^w)) > \tau), \quad (2)$$

where $\hat{\mathcal{F}}(x_i^w) \in \mathbb{R}^{K \times HW}$ is the logits output produced by the semantic segmentation network $\mathcal{F}$ and K is the class number. τ is a threshold used to screen high-confidence predicted pixels as the pseudo label.

However, $\mathcal{L}_u^h$ only treats $\mathcal{F}(x_i^w)$ as the hard pseudo label for $\mathcal{F}(x_i^s)$ and thus ignores additional information stored in logits $\hat{\mathcal{F}}(x_i^w)$. Taking this into account, we further consider the consistency between the logits of the weakly and strongly augmented images in high-confidence regions. In Eqn. (3), we give the formula of $\mathcal{L}_u^s$ for soft supervision.

$$\mathcal{L}_u^s = \frac{1}{N} \sum_{i=1}^N \text{KL}(\hat{\mathcal{F}}(x_i^s), \hat{\mathcal{F}}(x_i^w)) \odot \mathcal{M}_i, \quad (3)$$

where $\text{KL}(\cdot)$ is Kullback-Leibler Divergence loss function. We view the above framework as our baseline.

3.2. Pixel Propagation

As discussed in Sec. 1, pseudo labels obtained through threshold-based selection overlook the semantic similarity between pixels, constraining the utilization of unlabeled data. In this section, we propose the pixel propagation strategy to enhance the model’s overall awareness of pairwise similarities and consequently improve the utilization of unlabeled data, which involves two steps: (1) calculating correlation maps and (2) spreading correlation maps into predictions.

We first extract features w_1 and $w_2 \in \mathbb{R}^{D \times HW}$ through linear layers after the encoder of the network, where D is the channel dimension and HW is the number of feature vectors. These extracted features enable correlation matching to quantify the degree of pairwise similarity. Thus, we compute the correlation map $\mathcal{C}$ by performing a matrix multiplication between all pairs of feature vectors:

$$\mathcal{C} = \text{Softmax}(w_1^\top \cdot w_2) / \sqrt{D}, \quad (4)$$

where $^\top$ denotes the matrix transpose operation. The correlation map $\mathcal{C} \in \mathbb{R}^{HW \times HW}$ is a 2D matrix and is activated by a Softmax function to yield pairwise similarities. $\mathcal{C}$ enables accurate delineation of the corresponding regions belonging to the same object as shown in Fig. 2 and inspires us to propagate it into pseudo labels using correlation matching. More visualizations can be found in Fig. 3.

To enhance the model’s awareness of pairwise similarity, we spread the correlation map $\mathcal{C}$ into model logits outputs $\hat{\mathcal{F}}(x_i^u)$ to attain another representation of the prediction $\mathbf{z}_i^u \in \mathbb{R}^{K \times HW}$ via label propagation:

$$\mathbf{z}_i^u = f_1(\hat{\mathcal{F}}(x_i^u)) \cdot \mathcal{C}, \quad (5)$$

where $f_1(\cdot)$ is a bilinear interpolation for shape matching. The resulting $\mathbf{z}_i^u$ emphasizes the pairwise similarities of the same object through the correlation map.

Therefore, a correlation loss $\mathcal{L}_u^c$ can be calculated between $\mathbf{z}_i^u$ and the high-confidence pseudo labels as the supervision, which can be written as follows:

$$\mathcal{L}_u^c = \frac{1}{|N|} \sum_{i=1}^N (\ell_c(\mathbf{z}_i^u, \mathcal{F}(x_i^w))) \odot \mathcal{M}_i. \quad (6)$$

For the labeled images $\{x_i^l\}$, we also compute the cross-entropy loss between $\mathbf{z}_i^l$ and y_i^l as the supervised correlation loss $\mathcal{L}_s^c$, where $\mathbf{z}_i^l$ can be attained using Eqn. (5). So far, given a weakly augmented unlabeled image x_i^w , its correlation map $\mathcal{C}_i^w$ can effectively model pairwise similarities.

3.3. Region Propagation

During experiments, we also observe that every row $\mathbf{c}$ in $\mathcal{C}_i^w$ denotes the similarity between individual feature vectors and all vectors within the entire feature map, which implicitly encapsulates shape information. With this observation, we propose the region propagation strategy to enhance pseudo labels with these shapes information. Specifically, we first normalize $\mathbf{c}$ and turn it into a binary map $\hat{\mathbf{c}}$:

$$\hat{\mathbf{c}} = f_2(\mathbb{1}(\frac{\mathbf{c} - \min(\mathbf{c})}{\max(\mathbf{c}) - \min(\mathbf{c})} > 0.5)), \quad (7)$$

where $f_2(\cdot)$ is a shape-matching function to align the shapes of $\hat{\mathbf{c}}$ and $\mathcal{F}(x_i^w)$. As shown in Fig. 3, the shape information $\hat{\mathbf{c}} \in \mathbb{R}^{H \times W}$ explicitly embeds class agnostic shape information. For every $\hat{\mathbf{c}}$, we can calculate the overlap ratio r_1 between $\hat{\mathbf{c}}$ and the high-confidence regions $\mathcal{M}_i$. When $\hat{\mathbf{c}}$ has a large overlap with $\mathcal{M}_i$, (i.e., $r_1 > \tau$), we are able to use $\hat{\mathbf{c}}$ to adjust the pseudo label $\mathcal{F}(x_i^w)$.

Given the current pseudo labels $\mathcal{F}(x_i^w)$, we can calculate the quantity of each unique class $l \in L$ within high-confidence shape ($\mathcal{F}(x_i^w) \odot \mathcal{M}_i \odot \hat{\mathbf{c}}$) by a function $G(l)$ and locate the most significant class k^* with the following equation:

$$k^* = \operatorname{argmax}_{l \in L} G(l), \quad (8)$$

$$G(l) = \sum_{HW} \mathbb{1}[(\mathcal{F}(x_i^w) \odot \mathcal{M}_i \odot \hat{\mathbf{c}}) = l], \quad (9)$$

where L is the set of all unique classes that present in predictions $\mathcal{F}(x_i^w)$. With the most significant class k^* , we can calculate its proportion r_2 within the high-confidence shape.

When k^* highly coincides with the high-confidence shape, (i.e., $r_2 > \tau$), we can propagate the specific class k^* into the enhanced pseudo label $\mathcal{F}(x_i^w)$ and expanded high-confidence regions $\mathcal{M}_i$ by matching the certain shape $\hat{\mathbf{c}}$.

$$\mathcal{F}(x_i^w) = \begin{cases} k^*, & \hat{\mathbf{c}} = 1 \\ \mathcal{F}(x_i^w), & \hat{\mathbf{c}} = 0 \end{cases}, \mathcal{M}_i = \mathcal{M}_i \cup \hat{\mathbf{c}} \quad (10)$$

However, considering the intricate computations required for each specific shape within the correlation map and the frequent occurrence of similar semantic information in adjacent

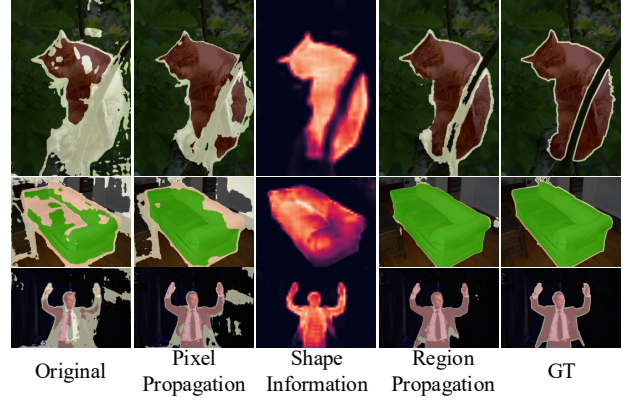


Figure 3. Illustration of our proposed propagation strategies. White areas are ignored regions due to low confidence. Combining the shape information with the most salient class, CorrMatch can significantly enhance pseudo labels and expand high-confidence regions.

regions, resulting in similar shapes in the correlation map, it becomes evident that involving every row of the correlation map in pseudo labels optimization is redundant. Hence, we employed a random sampling approach within the correlation map to expedite label propagation. As shown in Fig. 3, region propagation significantly expands high-confidence regions with shape information and the most salient class.

It is also worth mentioning that the correlation map construction process and label propagation only participate in the training process and hence do not bring any additional computational burdens during the inference process.

3.4. More Details

Dynamic threshold. As mentioned in FreeMatch [51], using a fixed threshold τ that is too strict or too loose is detrimental to model convergence. At the same time, we observe that the most suitable thresholds are different for different experimental settings (Fig. 5d). Thus, We provide a dynamic threshold strategy that is related to the training process.

Given the threshold τ a relatively small value (0.85) as initialization, the strategy of updating τ depends on the logits $\hat{\mathcal{F}}(x_i^w)$. We use the exponential moving average (EMA) [42] to iteratively update τ . Each increment is defined as:

$$\Delta\tau = \frac{1}{|L|} \sum_{l \in L} \max[\mathbb{1}(\mathcal{F}(x_i^w) = l) \odot \max(\hat{\mathcal{F}}(x_i^w))], \quad (11)$$

where $\max(\cdot)$ denotes taking the maximum value along the channel dimension. This operation aims to take the maximum confidence of all predicted classes in $\hat{\mathcal{F}}(x_i^w)$ and use their average as the increment for each iteration. We found that such a simple threshold updating strategy works well. We will further show in Sec. 4.3 that τ is insensitive to initialization. The corresponding pseudo code is provided in the supplementary materials.

Table 2. Comparisons of CorrMatch with the state-of-the-art approaches on the Pascal VOC 2012 val set in terms of mIoU (%). All methods are trained on the classic setting, *i.e.*, the labeled images are selected from the original VOC train set, which consists of 1,464 images.

Method	Training Size	1/16 (92)	1/8 (183)	1/4 (366)	1/2 (732)	Full (1464)
ST++ [64]	321 × 321	65.2	71.0	74.6	77.3	79.1
UniMatch [63]	321 × 321	75.2	77.2	78.8	79.9	81.2
Mean Teacher [48]	513 × 513	51.7	58.9	63.9	69.5	71.0
CutMix-Seg [11]	513 × 513	52.2	63.5	69.5	73.7	76.5
PseudoSeg [78]	513 × 513	57.6	65.5	69.1	72.4	73.2
CPS [6]	513 × 513	64.1	67.4	71.7	75.9	-
PC ² Seg [74]	513 × 513	57.0	66.3	69.8	73.1	74.2
U ² PL [52]	513 × 513	68.0	69.2	73.7	76.2	79.5
PS-MT [39]	513 × 513	65.8	69.6	76.6	78.4	80.0
GTA [29]	513 × 513	70.0	73.2	75.6	78.4	80.5
PCR [61]	513 × 513	70.1	74.7	77.2	78.5	80.7
RC ² L [70]	513 × 513	65.3	68.9	72.2	77.1	79.3
CCVC [54]	513 × 513	70.2	74.4	77.4	79.1	80.5
CorrMatch	321 × 321	76.4	78.5	79.4	80.6	81.8

Loss function. The overall objective function $\mathcal{L}$ is a combination of supervised loss $\mathcal{L}_s$ and unsupervised loss $\mathcal{L}_u$: $\mathcal{L} = \frac{1}{2}(\mathcal{L}_s + \mathcal{L}_u)$. Like most methods, we use the cross-entropy loss function $\mathcal{L}_s^h$ as the basic supervision of labeled data $\mathcal{D}^l$. Therefore, the supervised loss $\mathcal{L}_s$ is defined as the combination of $\mathcal{L}_s^h$ and supervised correlation loss $\mathcal{L}_s^c$: $\mathcal{L}_s = \frac{1}{2}(\mathcal{L}_s^h + \mathcal{L}_s^c)$. As for unsupervised loss $\mathcal{L}_u$ on unlabeled data $\mathcal{D}^u$, it can be expressed as follows:

$$\mathcal{L}_u = \lambda_1 \mathcal{L}_u^h + \lambda_2 \mathcal{L}_u^s + \lambda_3 \mathcal{L}_u^c, \quad (12)$$

where $\mathcal{L}_u^h$, $\mathcal{L}_u^s$ and $\mathcal{L}_u^c$ denote the unsupervised hard loss, soft loss, and correlation loss. And $[\lambda_1, \lambda_2, \lambda_3]$ are set to $[0.5, 0.25, 0.25]$ by default.

4. Experiments

4.1. Experiment Setup

Datasets. We report results on the Pascal VOC 2012 and Cityscapes datasets. Pascal VOC 2012 is a semantic segmentation benchmark with 21 classes, consisting of 1,464 high-quality annotated images for training and 1,449 images for evaluation originally [10]. We also conduct experiments on the aug Pascal VOC 2012 dataset, which contains more coarsely annotated images from the Segmentation Boundary Dataset (SBD) [20], resulting in 10,582 training images in total. Cityscapes is an urban scene understanding dataset, including 2,975 training and 500 validation images with fine annotations [7]. It contains 19 classes of urban scenes, and all images have the resolution of 1024×2048 .

Implementation details. Following most previous semi-supervised semantic segmentation methods, we use DeepLabV3+ [5] with ResNet-101 [21] pre-trained on ImageNet [8] as the backbone. For the training on the Pascal VOC 2012 dataset, we use stochastic gradient descent (SGD)

optimizer with an initial learning rate of 0.001, weight decay of $1e-4$, crop size of 321×321 or 513×513 , batch size of 16, and training epochs of 80. For the Cityscapes dataset, following UniMatch [63], we use stochastic gradient descent (SGD) optimizer with an initial learning rate of 0.005, weight decay of $1e-4$, crop size of 801×801 , batch size of 16, and training epochs of 240 with $4 \times$ A40 GPUs.

As for evaluation metrics, we report the mean Intersection-over-Union (mIoU) with original images following previous papers [6, 11, 39] for the Pascal VOC 2012 dataset. For Cityscapes, same as previous methods [6, 52, 63], we apply slide window evaluation with a fixed crop in a sliding window manner and then calculate mIoU on these cropped images. All the results are measured on the standard validation set based on single-scale inference.

4.2. Comparison with State-of-the-art Methods

Results on classic Pascal VOC 2012. We show the performance of our method with other state-of-the-art methods on the classic Pascal VOC 2012 Dataset in Tab. 2. Our experiments are conducted on various splits of the original train set following the data partition in CPS [6]. On the full split, our method gets 81.8% mIoU. Also, CorrMatch achieves consistent performance gains compared to existing state-of-art approaches. Particularly, CorrMatch outperforms UniMatch by 1.2%, 1.3%, 0.6%, 0.7% and 0.6% on each split.

Results on aug Pascal VOC 2012. In Tab. 3, we show our performance and compare it with existing methods on the aug Pascal VOC 2012 Dataset. It is clear that our results are consistently much better than the existing best ones. We conduct experiments on 1/16, 1/8, and 1/4 splits, respectively. Under the 321×321 training size (top-left of Tab. 3), compared to the supervised baseline, CorrMatch gets +12.0%, +7.4%, and +5.5% improvements. In addition, our approach

Table 3. Comparisons of state-of-the-art methods on the Pascal VOC 2012 val set with mIoU (%) metric. All methods are trained on the aug setting, i.e., the labeled images are selected from the aug VOC train set, which consists of 10, 582 images. [†] means using U²PL [52]’s splits.

Method	Train size	1/16 (662)	1/8 (1323)	1/4 (2646)
Supervised	321 × 321	65.6	70.4	72.8
ST++ [64]	321 × 321	74.5	76.3	76.6
CAC [35]	321 × 321	72.4	74.6	76.3
UniMatch [63]	321 × 321	76.5	77.0	77.2
CorrMatch	321 × 321	77.6	77.8	78.3
U2PL [†] [52]	513 × 513	77.2	79.0	79.3
GTA [†] [29]	513 × 513	77.8	80.4	80.5
PCR [†] [61]	513 × 513	78.6	80.7	80.7
CCVC [†] [61]	513 × 513	76.8	79.4	79.6
AugSeg [†] [73]	513 × 513	79.3	81.5	80.5
CorrMatch[†]	513 × 513	81.3	81.9	80.9

Method	Train size	1/16 (662)	1/8 (1323)	1/4 (2646)
CutMix-Seg [11]	513 × 513	71.7	75.5	77.3
CCT [41]	513 × 513	71.9	73.7	76.5
GCT [32]	513 × 513	70.9	73.3	76.7
CPS [6]	513 × 513	74.5	76.4	77.7
AEL [23]	513 × 513	77.2	77.6	78.1
FST [9]	513 × 513	73.9	76.1	78.1
ELN [34]	513 × 513	-	75.1	76.6
U ² PL [52]	513 × 513	74.4	77.6	78.7
PS-MT [39]	513 × 513	75.5	78.2	78.7
AugSeg [73]	513 × 513	77.0	77.3	78.8
CorrMatch	513 × 513	78.4	79.3	79.6

outperforms UniMatch by 1.1%, 0.8%, and 1.1% on each split. As for the 513×513 training size (right of Tab. 3), CorrMatch also consistently outperforms current state-of-the-art methods. For instance, we get 79.3% mIoU on the 1/8 split with a gain of around 2% compared to AugSeg [73].

We also report the results using the same splits as in U²PL [52] with 513×513 training size (bottom-left of Tab. 3), which contain more well-annotated labels and have higher expectations of results. Compared to the best method AugSeg [73], our method gains 2.0% improvement on the 1/16 split. Furthermore, same to other methods, we observe that, as the split size increases from 1/8 to 1/4, the performance decreases under this setting. This is because in the 1/8 split, almost all of the accurately labeled images are included, and most of the images added to the larger split are coarsely labeled, which results in no improvement in performance.

Results on Cityscapes. In Tab. 4, we compare the performance of CorrMatch with state-of-the-art methods on the Cityscapes dataset. We follow sliding window evaluation and online hard example mining (OHEM) loss [45] techniques, which have been widely applied in previous SOTA works [6, 23, 39, 52, 61, 63]. It can be clearly seen that our method can consistently outperform other methods under all splits. Compared to UniMatch [63], our CorrMatch achieves +0.7%, +0.6%, +0.2%, and +0.9% on 1/16, 1/8, 1/4, 1/2 splits, respectively.

4.3. Ablations Studies

In this part, we conduct a series of ablations studies to verify the designs of proposed strategies in CorrMatch. We report the results of the DeepLabV3+ network using ResNet-101 as the encoder on the original Pascal VOC 2012 dataset with training size 321 × 321.

Effectiveness of components. We first conduct ablation studies on different components of our CorrMatch to demon-

Table 4. Comparing results of state-of-the-art algorithms on the Cityscapes val set. All the experiments are conducted with ResNet-101 as the backbone.

Method	1/16 (186)	1/8 (372)	1/4 (744)	1/2 (1488)
Supervised	65.7	72.5	74.4	77.8
CCT [41]	69.3	74.1	76.0	78.1
CPS [6]	69.8	74.3	74.6	76.8
AEL [23]	74.5	75.5	77.5	79.0
U ² PL [52]	70.3	74.4	76.5	79.1
PS-MT [39]	-	76.9	77.6	79.1
UniMatch [63]	76.6	77.9	79.2	79.5
PCR [61]	73.4	76.3	78.4	79.1
CorrMatch	77.3	78.5	79.4	80.4

strate their effectiveness in Tab. 5. With the hard unsupervised loss and dynamic threshold, we get 73.6% on the 92 split and 80.0% on the 1464 split. Adding soft loss $\mathcal{L}_u^s$ as the basic framework brings 0.8% and 0.5% improvements. With the help of label propagation, we achieve another 2.0% and 1.3% improvements. These results demonstrate the effectiveness of each of our components individually. Also, replacing $\mathcal{L}_u^h$ with $\mathcal{L}_u^s$ results in a performance decrease, which illustrates the importance of $\mathcal{L}_u^h$. Finally, the complete CorrMatch achieves 76.4% and 81.8% mIoU, which is +2.8% and +1.8% compared to the baselines.

We also conduct experiments with the fixed threshold (0.95). It can be observed that compared to the fixed baselines (73.1% and 79.9%), changing it into a dynamic manner only brings +0.5% and +0.1%. Meanwhile, after adding all components, the corresponding improvements can be lifted to +0.9% and +1.0%. This proves our threshold strategy cooperates well with our label propagation strategy.

Impact of label propagation strategies. In Tab. 6, we conduct the ablation study of our label propagation strategies. Our pixel propagation strategy, which constructs the cor-

Table 5. Ablation study on the effectiveness of different components, including threshold τ (Dyna. denotes our dynamic strategy), hard loss $\mathcal{L}_u^h$, soft loss $\mathcal{L}_u^s$, label propagation $\mathcal{P}$.

τ	$\mathcal{L}_u^h$	$\mathcal{L}_u^s$	$\mathcal{P}$	92	1464
Dyna.	✓			73.6	80.0
Dyna.		✓		73.1	79.6
Dyna.	✓	✓		74.4	80.5
Dyna.	✓		✓	74.6	80.6
Dyna.	✓	✓	✓	76.4	81.8
Fixed	✓			73.1	79.9
Fixed	✓	✓		73.3	79.9
Fixed	✓		✓	74.3	80.1
Fixed	✓	✓	✓	75.5	80.8

Table 6. Ablation study on the label propagation strategies.

Method	92	366	1464
w/o Propagation	74.4	78.5	80.5
w/ Pixel Propagation	75.8	78.9	81.3
w/ Pixel & Region Propagation	76.4	79.4	81.8

relation maps and spreads them into predictions as a new representation with the supervision of correlation loss $\mathcal{L}^c$, brings 1.4%, 0.4%, and 0.8% improvements. Furthermore, equipped with our region propagation strategy, more detailed local shape information is mined and thus enhanced pseudo labels are obtained. This strategy further improves 0.6%, 0.5%, and 0.5% on 92, 366, and 1464 splits, respectively.

Where to extract features. In the default setting, we choose to extract features from the backbone, which makes the proposed strategies more convenient to be transplanted to other segmentation networks. Actually, given a specific network structure, the position of feature extraction can be flexible. Here, we consider the impact of different feature extraction positions on performance. In Tab. 7, we demonstrate the performance of extracting features after different positions for the Deeplabv3+ decoder under different splits. The results show that using the backbone features consistently outperforms other alternatives.

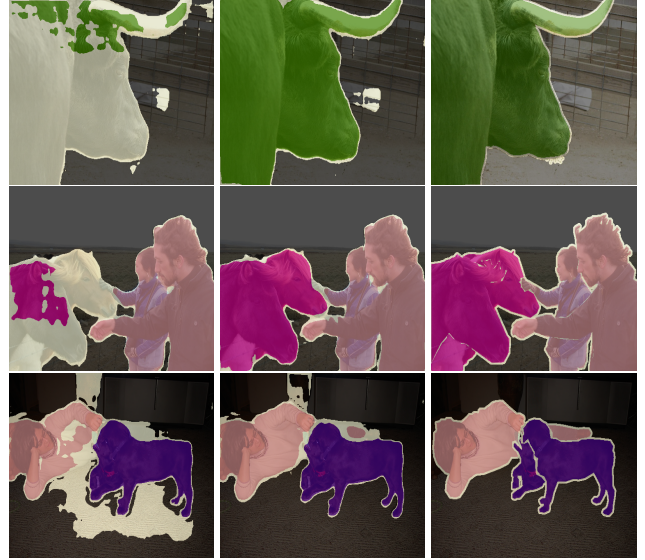
Different sampling strategies. Since using all shapes within the correlation map to enhance pseudo labels would incur a substantial computational burden, it is imperative to sample a subset of shapes from it. Here we conduct experiments about sampling methods and quantities in Tab. 8. We conduct experiments on random sampling $\mathcal{R}$ and uniform sampling $\mathcal{U}$ methods, with 16, 32, 64, 128, and 256 sampling numbers on the 1464 split. The results show random sampling continuously outperforms uniform sampling. Among these, random sampling with 128 sample numbers yields the best performance, with marginal differences compared to the 256-sample strategy. Thus, we choose to randomly sample 128 shapes from the correlation map as a trade-off between

Table 7. Ablation study on feature extraction positions. We take features after each specific module of DeepLabV3+ to build correlation maps and adopt label propagation strategies.

Position	Backbone	ASPP	Fusion	Classifier
732	80.4	79.5	79.1	79.5
1464	81.8	80.6	80.1	80.8

Table 8. Ablation study on the different sampling methods. $\mathcal{R}$ denotes random sampling; $\mathcal{U}$ denotes uniform sampling.

Numbers	16	32	64	128	256
$\mathcal{R}$	81.1	81.2	81.4	81.8	81.7
$\mathcal{U}$	81.0	81.1	81.2	81.4	81.0



(a) w/o propagation (b) w/ propagation (c) GT

Figure 4. Qualitative results on the Pascal VOC 2012 dataset. (a) Pseudo labels without label propagation; (b) Pseudo labels with CorrMatch; (c) Ground truth. White areas in (a) and (b) are ignored regions due to low confidence.

computational efficiency and performance.

Different initial values for CorrMatch. Since our EMA-based threshold updating strategy needs an initial value for τ , we discuss the impact of different initialization values for τ in Fig. 5a. The conclusion is that our threshold strategy is insensitive to different initialization values. Even with different threshold initialization values, all the thresholds tend to approach a similar value very quickly (around 1500 iterations) in the early stage of training (around 40000 iterations in total) under all experiment settings.

4.4. Correlation Helps Mining Reliable Regions

Statistics. Ideally, all correctly predicted points should be regarded as pseudo labels for the unlabeled data. To demonstrate the ability of correlation matching to help label propa-

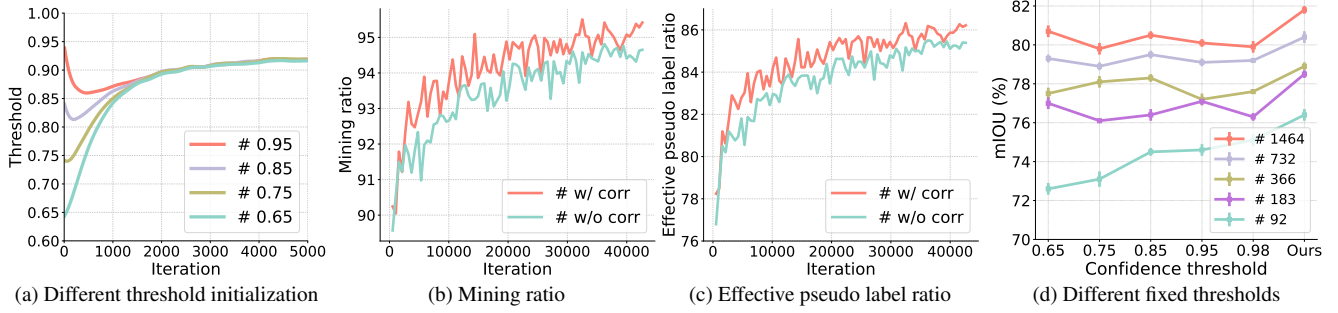


Figure 5. Some statistics on label propagation and the threshold strategy. For (a), (b), and (c), experiments are conducted on the 1464 split.

gation, we count the mining ratio and effective pseudo label ratio in Fig. 5b and Fig. 5c. The mining ratio is the proportion of selected high-confidence pixels among all correctly predicted pixels. The effective pseudo label ratio is the proportion of accurately predicted pseudo labels to the whole image, which can reflect effective pseudo label numbers. It can be clearly seen that with the proposed label propagation strategies, the mining ratio and effective pseudo label ratio are significantly higher than those without them, which illustrates that the utilization of unlabeled data has improved effectively. This further indicates our strategies can improve the overall quality of pseudo labels by leveraging similarity and shape information from correlation maps.

Qualitative analysis. In Fig. 4, we give some visualization results to further demonstrate the effectiveness of our label propagation strategies. Comparing Fig. 4b and Fig. 4a, it is obvious that with the support of label propagation, the number of pixels and completeness of the high-confidence regions are significantly better than those without it. This means that our method can effectively expand high-confidence regions and populate these regions with the correct categories. We will provide more detailed qualitative results in the supplementary materials.

5. Discussions on Label Propagation Strategy

Traditionally, semi-supervised semantic segmentation methods mostly rely on adjusting thresholds to expand high-confidence regions [52, 63]. However, selecting the most suitable threshold could be a challenging task. For instance, our observations illustrated in Fig. 5d, indicate that the optimal threshold can vary significantly. Fig. 6a and Fig. 6b further demonstrate that a too-strict threshold restricts the unlabeled data utilization, while a lenient threshold results in fragmented incorrect pixel predictions.

Different from the scheme of directly adjusting the threshold, label propagation does not merely expand the high-confidence regions; it assigns accurate predictions to pseudo labels by utilizing accurate shapes within the correlation map, which helps maintain more consistent semantic structures within high-confidence regions and thus mitigates the

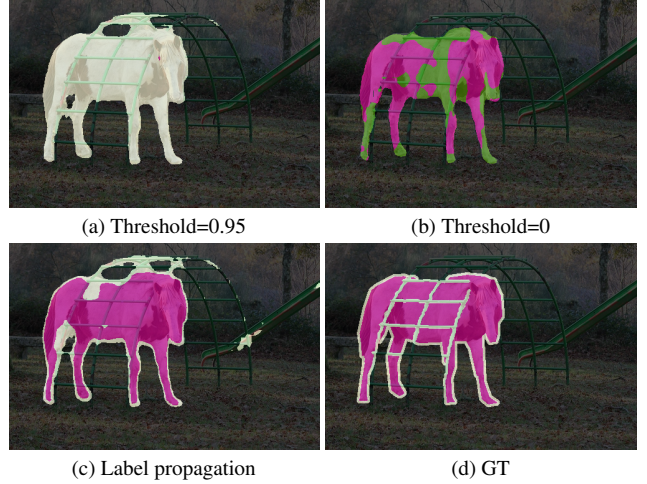


Figure 6. Comparisons of pseudo labels with different strategies. discontinuity issue. In Fig. 6c and the last column of Fig. 5d, we show the pseudo label and performance of CorrMatch. This indicates that our CorrMatch consistently obtains more accurate and complete pseudo labels and achieves the highest results on all splits.

6. Conclusions

We present CorrMatch that can utilize label propagation with correlation matching to discover more accurate high-confidence regions for semi-supervised semantic segmentation. The key contributions of our CorrMatch are reconsidering the use of correlation maps and designing two label propagation strategies to enrich the pseudo label. Equipped with these strategies, CorrMatch significantly expands the high-confidence regions and thus can utilize unlabeled data more efficiently. Experiments show the superiority of our CorrMatch over other methods.

Acknowledgments. This research was supported by NSFC (NO. 62225604, No. 62276145), the Fundamental Research Funds for the Central Universities (Nankai University, 070-63223049), CAST through Young Elite Scientist Sponsorship Program (No. YESS20210377). Computations were supported by the Supercomputing Center of Nankai University (NKSC).

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